

SAFETY DATA SHEET according to Regulation (EC) No. 1907/2006

TARTARIC ACID (L) CRYSTALS (E334) PHARM.

Version 2.0

Print Date 26.07.2023

Revision date / valid from 24.07.2023

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Trade name : TARTARIC ACID (L) CRYSTALS (E334) PHARM.
Substance name : (+)-Tartaric acid
CAS-No. : 87-69-4
EC-No. : 201-766-0
EU REACH-Reg. No. : 01-2119537204-47-xxxx

1.2. Relevant identified uses of the substance or mixture and uses advised against

Use of the Substance/Mixture : Identified use: See table in front of appendix for a complete overview of identified uses.
Uses advised against : At this moment we have not identified any uses advised against
Remarks : Before referring to any Exposure Scenario attached to this Safety Data Sheet please check the grade of the product: the Exposure Scenarios presented are not related to all product grade

1.3. Details of the supplier of the safety data sheet

Company : Brenntag N.V.
Nijverheidslaan 38
BE 8540 Deerlijk
Telephone : +32 (0)56 77 6944
Telefax : +32 (0)56 77 5711
E-mail address : info@brenntag.be
Responsible/issuing person : Master Data Administration

Company : Brenntag Nederland B.V.
Donker Duyvisweg 44
NL 3316 BM Dordrecht
Telephone : +31 (0)78 65 44 944
Telefax : +31 (0)78 65 44 919
E-mail address : info@brenntag.nl
Responsible/issuing person : Master Data Administration

1.4. Emergency telephone number

Emergency telephone number : Belgium: Antipoison Center - Brussels TEL: +32(0)70 245 245

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Netherland: National Poisoning Information Center - Bilthoven
TEL: +31(0) 88 755 8000 (Only for the purpose of informing
medical personnel in cases of acute intoxications)

SECTION 2: Hazards identification

2.1. Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008

REGULATION (EC) No 1272/2008			
Hazard class	Hazard category	Target Organs	Hazard statements
Serious eye damage	Category 1	---	H318

For the full text of the H-Statements mentioned in this Section, see Section 16.

Most important adverse effects

Human Health : See section 11 for toxicological information.
Physical and chemical hazards : See section 9/10 for physicochemical information.
Potential environmental effects : See section 12 for environmental information.

2.2. Label elements

Labelling according to Regulation (EC) No 1272/2008

Hazard symbols :



Signal word : Danger

Hazard statements : H318 Causes serious eye damage.

Precautionary statements

Prevention : P280 Wear eye protection/ face protection.

Response : P305 + P351 + P338 + P310 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/doctor.

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Hazardous components which must be listed on the label:

- (+)-Tartaric acid

2.3. Other hazards

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

|| Ecological information: No information available about endocrine disruption properties for environment.

|| Toxicological information: No information available about endocrine disruption properties for human health.

|| Risk of dust explosion.

SECTION 3: Composition/information on ingredients

3.1. Substances

		Classification (REGULATION (EC) No 1272/2008)	
Hazardous components	Amount [%]	Hazard class / Hazard category	Hazard statements
(+)-Tartaric acid			
CAS-No. : 87-69-4	100	Eye Dam.1	H318
EC-No. : 201-766-0			
EU REACH- : 01-2119537204-47-xxxx			
Reg. No.			

For the full text of the H-Statements mentioned in this Section, see Section 16.

SECTION 4: First aid measures

4.1. Description of first aid measures

- General advice : Move out of dangerous area. Take off all contaminated clothing immediately.
- If inhaled : Remove to fresh air. If symptoms persist, call a physician.
- In case of skin contact : Wash off immediately with soap and plenty of water. If skin

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	irritation persists, call a physician.
In case of eye contact	: Rinse immediately with plenty of water, also under the eyelids, for at least 10 minutes. Consult an eye specialist immediately. Go to an ophthalmic hospital if possible.
If swallowed	: Rinse mouth with water. Never give anything by mouth to an unconscious person. If symptoms persist, call a physician.
Protection of First Aid Responders	: First Aid responders should pay attention to self-protection and use the recommended protective clothing.

4.2. Most important symptoms and effects, both acute and delayed

Symptoms	: See Section 11 for more detailed information on health effects and symptoms.
Effects	: May cause irreversible eye damage. See Section 11 for more detailed information on health effects and symptoms.

4.3. Indication of any immediate medical attention and special treatment needed

Treatment	: In case of accidental inhalation apply glucocorticoid inhalation spray. Treat symptomatically.
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SECTION 5: Firefighting measures

5.1. Extinguishing media

Suitable extinguishing media	: Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.
Unsuitable extinguishing media	: High volume water jet

5.2. Special hazards arising from the substance or mixture

Specific hazards during firefighting	: In case of fire hazardous decomposition products may be produced such as:
Hazardous combustion products	: Carbon monoxide, Carbon dioxide (CO ₂)

5.3. Advice for firefighters

Special protective equipment for firefighters	: In the event of fire, wear self-contained breathing apparatus. Wear personal protective equipment.
Further advice	: Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Cool closed containers exposed to fire with water spray.

SECTION 6: Accidental release measures

TARTARIC ACID (L) CRYSTALS (E334) PHARM.**6.1. Personal precautions, protective equipment and emergency procedures**

Personal precautions : Use personal protective equipment. Keep away unprotected persons. Avoid dust formation. Ensure adequate ventilation. Avoid contact with skin and eyes.

6.2. Environmental precautions

Environmental precautions : Do not flush into surface water or sanitary sewer system. Avoid subsoil penetration. If the product contaminates rivers and lakes or drains inform respective authorities. If material reaches soil inform authorities responsible for such cases.

6.3. Methods and materials for containment and cleaning up

Methods and materials for containment and cleaning up : Use mechanical handling equipment. Keep in suitable, closed containers for disposal.

Further information : Treat recovered material as described in the section "Disposal considerations".

6.4. Reference to other sections

See Section 1 for emergency contact information.
See Section 8 for information on personal protective equipment.
See Section 13 for waste treatment information.

SECTION 7: Handling and storage**7.1. Precautions for safe handling**

Advice on safe handling : Keep container tightly closed. Avoid dust formation. Ensure adequate ventilation. Wear personal protective equipment. Avoid contact with skin, eyes and clothing. Emergency eye wash fountains and emergency showers should be available in the immediate vicinity.

Hygiene measures : Keep away from food, drink and animal feedingstuffs. Smoking, eating and drinking should be prohibited in the application area. Wash hands before breaks and at the end of workday. Take off all contaminated clothing immediately.

7.2. Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers : Store in original container.

Advice on protection against fire and explosion : Normal measures for preventive fire protection.

Further information on storage conditions : Keep tightly closed in a dry and cool place.

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Advice on common storage : Keep away from food, drink and animal feedingstuffs.

7.3. Specific end use(s)

Specific use(s) : Identified use: See table in front of appendix for a complete overview of identified uses.

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

Other Occupational Exposure Limit Values

(Additional) Information : Contains no substances with occupational exposure limit values.
Contains no substances with occupational exposure limit values.

Component:	(+)-Tartaric acid	CAS-No. 87-69-4
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Derived No Effect Level (DNEL)/Derived Minimal Effect Level (DMEL)

DNEL		
Workers, Long-term - systemic effects, Skin contact	:	2,9 mg/kg bw/day
DNEL		
Workers, Long-term - systemic effects, Inhalation	:	5,2 mg/m ³
DNEL		
Consumers, Long-term - systemic effects, Skin contact	:	1,5 mg/kg bw/day
DNEL		
Consumers, Long-term - systemic effects, Inhalation	:	1,3 mg/m ³
DNEL		
Consumers, Long-term - systemic effects, Ingestion	:	8,1 mg/kg bw/day

Predicted No Effect Concentration (PNEC)

Fresh water	:	0,312 mg/l
Marine water	:	0,312 mg/l
Intermittent releases	:	0,514 mg/l
Sewage treatment plant (STP)	:	10 mg/l
Fresh water sediment	:	1,141 mg/kg d.w.

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Marine sediment	: 1,141 mg/kg d.w.
Soil	: 0,045 mg/kg d.w.

8.2. Exposure controls

Appropriate engineering controls

Refer to protective measures listed in sections 7 and 8.

Personal protective equipment

Respiratory protection

Advice : Required if dust is released
Respiratory protection complying with EN 141.
Recommended Filter type:
Particle filter:P2

Hand protection

Advice : Protective gloves complying with EN 374.
Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.
Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion, and the contact time.
Protective gloves should be replaced at first signs of wear.

Material : Natural Rubber
Break through time : ≥ 8 h
Glove thickness : 0,5 mm

Material : polychloroprene
Break through time : ≥ 8 h
Glove thickness : 0,5 mm

Material : Nitrile rubber
Break through time : ≥ 8 h
Glove thickness : 0,35 mm

Material : butyl-rubber
Break through time : ≥ 8 h
Glove thickness : 0,5 mm

Material : Fluorinated rubber
Break through time : ≥ 8 h
Glove thickness : 0,4 mm

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Material : Polyvinylchloride
Break through time : ≥ 8 h
Glove thickness : 0,5 mm

Eye protection

Advice : Safety goggles

Skin and body protection

Advice : Wear personal protective equipment.

Environmental exposure controls

General advice : Do not flush into surface water or sanitary sewer system.
Avoid subsoil penetration.
If the product contaminates rivers and lakes or drains inform respective authorities.
If material reaches soil inform authorities responsible for such cases.

SECTION 9: Physical and chemical properties**9.1 Information on basic physical and chemical properties**

Form : solid

Physical state : solid

Colour : white

Odour : odourless

Odour Threshold : Not applicable

Melting point/range : 169 °C (1013 hPa)

Boiling point/boiling range : 179,1 °C (1013 hPa)

Flammability (solid, gas) : Not applicable

Upper explosion limit / Upper flammability limit : No data available

Lower explosion limit / Lower flammability limit : No data available

Flash point : 210 °C

Auto-ignition temperature : 425 °C

Decomposition temperature : > 170 °C

Self-Accelerating : No data available

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decomposition temperature
(SADT)

pH : 2,2
(as aqueous solution)

Viscosity

Viscosity, dynamic : No data available

Viscosity, kinematic : No data available

Flow time : No data available

Solubility(ies)

Water solubility : 1390 g/l (20 °C)

Solubility in other solvents : No data available

Dissolution Rate : No data available

Partition coefficient: n-
octanol/water : log Pow: -1,91 (20 °C)

Dispersion Stability : No data available

Vapour pressure : < 5 Pa (20 °C)

Relative density : No data available

Density : 1,76 g/cm³ (20 °C)

Bulk density : No data available

Relative vapour density : No data available

Particle characteristics
No data available

9.2 Other information

Explosives : Product is not explosive.

Oxidizing properties : not oxidising

Evaporation rate : Not applicable

SECTION 10: Stability and reactivity**10.1. Reactivity**

Advice : No decomposition if stored and applied as directed.

10.2. Chemical stability

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Advice : Stable under recommended storage conditions.

10.3. Possibility of hazardous reactions

Hazardous reactions : Under normal conditions of storage and use, hazardous reactions will not occur.

10.4. Conditions to avoid

Conditions to avoid : Heat, flames and sparks. Exposure to moisture
Thermal decomposition : > 170 °C

10.5. Incompatible materials

Materials to avoid : fluorine, alkalis

10.6. Hazardous decomposition products

Hazardous decomposition products : Fire may cause evolution of: Carbon oxides

SECTION 11: Toxicological information

11.1. Information on the hazard classes within the meaning of Regulation (EC) No. 1272/2008

Component:	(+)-Tartaric acid	CAS-No. 87-69-4
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Acute toxicity

Oral

LD50 : > 2000 mg/kg (Rat)

Inhalation

Study scientifically not justified.

Dermal

LD50 : > 2000 mg/kg (Rat)

Irritation

Skin

Result : No skin irritation (OECD Test Guideline 404)

Eyes

Result : Risk of serious damage to eyes. (OECD Test Guideline 437)

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Result : not sensitizing (OECD Test Guideline 429)

CMR effects**CMR Properties**

Carcinogenicity : Did not show carcinogenic effects in animal experiments.
Mutagenicity : In vitro tests did not show mutagenic effects
In vivo tests did not show mutagenic effects
Teratogenicity : Did not show teratogenic effects in animal experiments.
Reproductive toxicity : Study scientifically not justified.

Specific Target Organ Toxicity**Single exposure**

Remarks : The substance or mixture is not classified as specific target organ toxicant, single exposure.

Repeated exposure

Remarks : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Other toxic properties**Repeated dose toxicity**

NOAEL : 2460 mg/kg bw/day
(Rat)(Oral; 2 y)

Aspiration hazard

No aspiration toxicity classification,

Further information

Other relevant toxicity information : Dust in the eyes may cause mechanical irritation.

11.2. Information on other hazards**Data for the product**

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Endocrine disrupting properties

	Assessment	:	No information available about endocrine disruption properties for human health.
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	Component:	(+)-Tartaric acid	CAS-No. 87-69-4
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Endocrine disrupting properties

	Assessment	:	No information available about endocrine disruption properties for human health.
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SECTION 12: Ecological information

12.1. Toxicity

	Component:	(+)-Tartaric acid	CAS-No. 87-69-4
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Acute toxicity

Fish

LC0	:	200 mg/l (Carassius auratus (goldfish))
LC50	:	> 100 mg/l (Fish; 96 h)

Toxicity to daphnia and other aquatic invertebrates

EC50	:	135 mg/l (Daphnia magna (Water flea); 24 h)
EC50	:	93,3 mg/l (Daphnia (water flea); 48 h)

algae

ErC50	:	51,4 mg/l (algae; 72 h)
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Bacteria

EC50	:	> 1000 mg/l (activated sludge; 3 h) (static test; OECD Test Guideline 209)
EC10	:	> 1000 mg/l (activated sludge; 3 h) (static test; OECD Test Guideline 209)

12.2. Persistence and degradability

	Component:	(+)-Tartaric acid	CAS-No. 87-69-4
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Persistence and degradability

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Persistence

Result : No data available

Biodegradability

Result : 85 % (Related to: O2 consumption; Exposure Time: 28 d)(OECD Test Guideline 306) Readily biodegradable.

12.3. Bioaccumulative potential

Component:	(+)-Tartaric acid	CAS-No. 87-69-4
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Bioaccumulation

Result : log Kow -1,91 (20 °C)
: Bioaccumulation is not expected.
log Pow < 1

12.4. Mobility in soil

Component:	(+)-Tartaric acid	CAS-No. 87-69-4
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Mobility

Water : The product is water soluble.

12.5. Results of PBT and vPvB assessment

Data for the product

Results of PBT and vPvB assessment

Result : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

Component:	(+)-Tartaric acid	CAS-No. 87-69-4
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Results of PBT and vPvB assessment

Result : This substance is not considered to be persistent, bioaccumulating nor toxic (PBT)., This substance is not considered to be very persistent and very bioaccumulating (vPvB).

12.6. Endocrine disrupting properties

Data for the product

Endocrine disrupting : No information available about endocrine disruption properties for

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potential environment.

Component:	(+)-Tartaric acid	CAS-No. 87-69-4
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Endocrine disrupting potential : No information available about endocrine disruption properties for environment.

12.7. Other adverse effects

Component:	(+)-Tartaric acid	CAS-No. 87-69-4
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Additional ecological information

Result : Do not flush into surface water or sanitary sewer system. Avoid subsoil penetration.

SECTION 13: Disposal considerations

13.1. Waste treatment methods

Product : Disposal together with normal waste is not allowed. Special disposal required according to local regulations. Do not let product enter drains. Contact waste disposal services. This product shall be disposed of or recovered in compliance with Directive 2008/98/EC on waste as lastly amended.

Contaminated packaging : Empty contaminated packagings thoroughly. They can be recycled after thorough and proper cleaning. If recycling is not practicable, dispose of in compliance with local regulations.

European Waste Catalogue Number : No waste code according to the European Waste Catalogue can be assigned for this product, as the intended use dictates the assignment. The waste code is established in consultation with the regional waste disposer.

SECTION 14: Transport information

Not dangerous goods for ADR, RID, IMDG and IATA.

14.1. UN number or ID number

Not applicable.

14.2. UN proper shipping name

Not applicable.

14.3. Transport hazard class(es)

Not applicable.

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14.4. Packaging group

Not applicable.

14.5. Environmental hazards

Not applicable.

14.6. Special precautions for user

Not applicable.

14.7 Maritime transport in bulk according to IMO instruments

Not applicable for product as supplied.

SECTION 15: Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Component:	(+)-Tartaric acid	CAS-No. 87-69-4
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EU. Chemicals Subject : ; The substance/mixture does not fall under this legislation.
to PIC Procedure:
Regulation 649/2012/EU
on export and import of
dangerous chemicals, as
amended

EU. REACH, Annex XVII, : ; The substance/mixture does not fall under this legislation.
Marketing and Use
Restrictions (Regulation
1907/2006/EC)

Component:	(+)-Tartaric acid	CAS-No. 87-69-4
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EU. Regulation : EC Number: , 201-766-0; Category 1 - Substances authorised
528/2012/EU concerning as food additives according to Regulation (EC) No 1333/2008;
the making available on Concentration to be limited so that each biocidal product does
the market and use of not require classification according to either Directive
biocidal products, Annex 1999/45/EC or Regulation (EC) No 1272/2008.
I: Active substances

Component:	(+)-Tartaric acid	CAS-No. 87-69-4
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EU. Directive : ; The substance/mixture does not fall under this legislation.
2012/18/EU (SEVESO
III) on major accident
hazards involving

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dangerous substances,
Annex I

Notification status (+)-Tartaric acid:

Regulatory List	Notification	Notification number
EINECS	YES	201-766-0
DSL	YES	
KECI (KR)	YES	KE-10801
ENCS (JP)	YES	(2)-1456
ISHL (JP)	YES	(2)-1456
INSQ	YES	
IECSC	YES	
ONT INV	YES	
TCSI	YES	
PICCS (PH)	YES	
TSCA	YES	
PHARM (JP)	YES	
VN INV	YES	
TH INV	YES	2918.12
TH INV	YES	55-1-06626
AU AIICL	YES	
NZIOC	YES	

15.2. Chemical safety assessment

A Chemical Safety Assessment has been carried out for this substance.

SECTION 16: Other information

Full text of H-Statements referred to under sections 2 and 3.

H318 Causes serious eye damage.

Abbreviations and Acronyms

AU AIICL	Australia. Industrial Chemicals Act (AII) List
BCF	bioconcentration factor
BOD	biochemical oxygen demand
CAS	Chemical Abstracts Service
CLP	Classification, Labelling and Packaging
CMR	carcinogenic, mutagenic or toxic to reproduction
COD	chemical oxygen demand
DNEL	derived no-effect level
DSL	Canada. Environmental Protection Act, Domestic Substances List
EINECS	European Inventory of Existing Commercial Chemical Substances
ELINCS	European List of Notified Chemical Substances

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ENCS (JP)	Japan. Kashin-Hou Law List
GHS	Globally Harmonized System of Classification and Labelling of Chemicals
IECSC	China. Inventory of Existing Chemical Substances
INSQ	Mexico. National Inventory of Chemical Substances
ISHL (JP)	Japan. Inventory of Industrial Safety & Health
KECI (KR)	Korea. Existing Chemicals Inventory
LC50	median lethal concentration
LOAEC	lowest observed adverse effect concentration
LOAEL	lowest observed adverse effect level
LOEL	lowest observed effect level
NDSL	Canada. Environmental Protection Act. Non-Domestic Substances List
NLP	no-longer polymer
NOAEC	no observed adverse effect concentration
NOAEL	no observed adverse effect level
NOEC	no observed effect concentration
NOEL	no observed effect level
NZIOC	New Zealand. Inventory of Chemicals
OECD	Organisation for Economic Cooperation and Development
OEL	occupational exposure limit
ONT INV	Canada. Ontario Inventory List
PBT	persistent, bioaccumulative and toxic
PHARM (JP)	Japan. Pharmacopoeia Listing
PICCS (PH)	Philippines. Inventory of Chemicals and Chemical Substances
PNEC	predicted no-effect concentration
REACH Auth. No.:	REACH Authorisation Number
REACH AuthAppC. No.	REACH Authorisation Application Consultation Number
UK REACH Auth. No.:	UK REACH Authorisation Number
UK REACH AuthAppC. No.	UK REACH Authorisation Application Consultation Number
UK REACH-Reg.No	UK REACH Registration Number
STOT	specific target organ toxicity
SVHC	substance of very high concern
TCSI	Taiwan. Existing Chemicals Inventory
TH INV	Thailand. Existing Chemicals Inventory from FDA
TSCA	US. Toxic Substances Control Act
UVCB	substance of unknown or variable composition, complex reaction products or biological materials
VN INVL	Vietnam. National Chemical Inventory
vPvB	very persistent and very bioaccumulative

TARTARIC ACID (L) CRYSTALS (E334) PHARM.**Further information**

- Key literature references and sources for data : Supplier information and data from the "Database of registered substances" of the European Chemicals Agency (ECHA) were used to create this safety data sheet.
- Methods used for product classification : The classification for human health, physical and chemical hazards and environmental hazards were derived from a combination of calculation methods and if available test data.
- Hints for trainings : The workers have to be trained regularly on the safe handling of the products based on the information provided in the Safety Data Sheet and the local conditions of the workplace. National regulations for the training of workers in the handling of hazardous materials must be adhered to.
- Other information :
The information provided in this Safety Data Sheet is correct to our knowledge at the date of its revision. The information given only describes the products with regard to safety arrangements and is not to be considered as a warranty or quality specification and does not constitute a legal relationship.
The information contained in this Safety Data Sheet relates only to the specific material designated and may not be valid for such material used in combination with any other material or in any process, unless specified in the text.

|| Indicates updated section.

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Version 1.0

Print Date 12.03.2013

Revision Date 12.03.2013

No.	Short title	Main User Group (SU)	Sector of Use (SU)	Product Category (PC)	Process Category (PROC)	Environmental Release Category (ERC)	Article Category (AC)	Specified
1	Manufacture of substance	3	8, 9	35, 39	1, 2, 3, 4, 8a, 8b, 9	1	4	ES10004
2	Use as an intermediate	3	8, 9	35, 39	1, 2, 3, 4, 8a, 8b, 9	6a, 6b	4	ES10264
3	Formulation & (re)packing of substances and mixtures	3	10	35, 39	5, 8a, 8b, 9	2	4	ES10006
4	Use in Cleaning Agents	21	NA	35	NA	8a	NA	ES10266
5	Use in road and construction applications	22	NA	NA	8a, 8b, 9	8c, 8f	NA	ES10008
6	Use in road and construction applications	21	NA	NA	NA	10a, 11a	4	ES10010
7	Use in glass and ceramics	22	NA	NA	8a, 8b, 9	8c, 8f	NA	ES10012
8	Use in glass and ceramics	21	NA	NA	NA	10a, 11a	4	ES10014

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1. Short title of Exposure Scenario 1: Manufacture of substance

Main User Groups	SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Sectors of end-use	SU8: Manufacture of bulk, large scale chemicals (including petroleum products) SU9: Manufacture of fine chemicals
Chemical product category	PC35: Washing and cleaning products (including solvent based products) PC39: Cosmetics, personal care products
Process categories	PROC1: Use in closed process, no likelihood of exposure PROC2: Use in closed, continuous process with occasional controlled exposure PROC3: Use in closed batch process (synthesis or formulation) PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises PROC8a: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)
Article categories	AC4: Stone, plaster, cement, glass and ceramic articles
Environmental Release Categories	ERC1: Manufacture of substances
Activity	Manufacture of the substance. Includes, material transfers, storage, maintenance and loading (including marine vessel/barge, road/rail car and bulk container), sampling

2.1 Contributing scenario controlling environmental exposure for: ERC1

No exposure assessment presented for the environment.

2.2 Contributing scenario controlling worker exposure for: PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC9

Product characteristics	Concentration of the Substance in Mixture/Article	Covers percentage substance in the product up to 100 % (unless stated differently).
	Physical Form (at time of use)	solid
	Vapour pressure	< 5 Pa
	Process Temperature	20 °C
Frequency and duration of use	Covers daily exposures up to 8 hours (unless stated differently).	
Technical conditions and measures to control dispersion from source towards the worker	Provide a good standard of general ventilation. Natural ventilation is from doors, windows etc. Controlled ventilation means air is supplied or removed by a powered fan.(PROC4)	
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training. (Efficiency: 90 %)(PROC4, PROC8a)	
	Wear suitable gloves tested to EN374. (Efficiency: 80 %)(PROC8b, PROC9)	

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Wear a respirator conforming to EN 140/143 with type P1 filter or better
(Efficiency: 80 %)(PROC8a, PROC8b, PROC9)

3. Exposure estimation and reference to its source

Workers

Predicted exposures are not expected to exceed the applicable exposure limits when the operational conditions/risk management measures given in section 2 are implemented. The ECETOC TRA tool has been used to estimate workplace exposures unless otherwise indicated.

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

Health

Where other Risk Management Measures/Operational Conditions are adopted, then users should ensure that risks are managed to at least equivalent levels.

For scaling see: <http://www.ecetoc.org/tra>

Only properly trained persons shall make use of scaling methods while checking whether the OC and RMM are within the boundaries set by the ES

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1. Short title of Exposure Scenario 2: Use as an intermediate

Main User Groups	SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Sectors of end-use	SU8: Manufacture of bulk, large scale chemicals (including petroleum products) SU9: Manufacture of fine chemicals
Chemical product category	PC35: Washing and cleaning products (including solvent based products) PC39: Cosmetics, personal care products
Process categories	PROC1: Use in closed process, no likelihood of exposure PROC2: Use in closed, continuous process with occasional controlled exposure PROC3: Use in closed batch process (synthesis or formulation) PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises PROC8a: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)
Article categories	AC4: Stone, plaster, cement, glass and ceramic articles
Environmental Release Categories	ERC6a: Industrial use resulting in manufacture of another substance (use of intermediates) ERC6b: Industrial use of reactive processing aids
Activity	Use as an intermediate. Includes, material transfers, storage, maintenance and loading (including marine vessel/barge, road/rail car and bulk container), sampling

2.1 Contributing scenario controlling environmental exposure for: ERC6a, ERC6b

No exposure assessment presented for the environment.

2.2 Contributing scenario controlling worker exposure for: PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC9

Product characteristics	Concentration of the Substance in Mixture/Article	Covers percentage substance in the product up to 100 % (unless stated differently).
	Physical Form (at time of use)	solid
	Vapour pressure	< 5 Pa
	Process Temperature	20 °C
Frequency and duration of use	Covers daily exposures up to 8 hours (unless stated differently).	
Technical conditions and measures to control dispersion from source towards the worker	Provide a good standard of general ventilation. Natural ventilation is from doors, windows etc. Controlled ventilation means air is supplied or removed by a powered fan.(PROC4)	
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training. (Efficiency: 90 %)(PROC4, PROC8a)	
	Wear suitable gloves tested to EN374. (Efficiency: 80 %)(PROC8b, PROC9)	

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Wear a respirator conforming to EN 140/143 with type P1 filter or better
(Efficiency: 80 %)(PROC8a, PROC8b, PROC9)

3. Exposure estimation and reference to its source

Workers

Predicted exposures are not expected to exceed the applicable exposure limits when the operational conditions/risk management measures given in section 2 are implemented. The ECETOC TRA tool has been used to estimate workplace exposures unless otherwise indicated.

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

Health

Where other Risk Management Measures/Operational Conditions are adopted, then users should ensure that risks are managed to at least equivalent levels.

For scaling see: <http://www.ecetoc.org/tra>

Only properly trained persons shall make use of scaling methods while checking whether the OC and RMM are within the boundaries set by the ES

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1. Short title of Exposure Scenario 3: Formulation & (re)packing of substances and mixtures

Main User Groups	SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Sectors of end-use	SU 10: Formulation [mixing] of preparations and/ or re-packaging (excluding alloys)
Chemical product category	PC35: Washing and cleaning products (including solvent based products) PC39: Cosmetics, personal care products
Process categories	PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/or significant contact) PROC8a: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)
Article categories	AC4: Stone, plaster, cement, glass and ceramic articles
Environmental Release Categories	ERC2: Formulation of preparations
Activity	Formulation, packing and re-packing of the substance and its mixtures in batch or continuous operations, including storage, materials transfers, mixing, large and small scale packing, sampling, maintenance.

2.1 Contributing scenario controlling environmental exposure for: ERC2

No exposure assessment presented for the environment.

2.2 Contributing scenario controlling worker exposure for: PROC5, PROC8a, PROC8b, PROC9

Product characteristics	Concentration of the Substance in Mixture/Article	Covers percentage substance in the product up to 100 % (unless stated differently).
	Physical Form (at time of use)	solid
	Vapour pressure	< 5 Pa
	Process Temperature	20 °C
Frequency and duration of use	Covers daily exposures up to 8 hours (unless stated differently).	
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training. (Efficiency: 90 %)(PROC5, PROC8a)	
	Wear suitable gloves tested to EN374. (Efficiency: 80 %)(PROC8b, PROC9)	
	Wear a respirator conforming to EN 140/143 with type P1 filter or better (Efficiency: 80 %)(PROC5, PROC8a, PROC8b, PROC9)	

3. Exposure estimation and reference to its source

Workers

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Predicted exposures are not expected to exceed the applicable exposure limits when the operational conditions/risk management measures given in section 2 are implemented. The ECETOC TRA tool has been used to estimate workplace exposures unless otherwise indicated.

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

Health

Where other Risk Management Measures/Operational Conditions are adopted, then users should ensure that risks are managed to at least equivalent levels.

For scaling see: <http://www.ecetoc.org/tra>

Only properly trained persons shall make use of scaling methods while checking whether the OC and RMM are within the boundaries set by the ES

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1. Short title of Exposure Scenario 4: Use in Cleaning Agents

Main User Groups	SU 21: Consumer uses: Private households (= general public = consumers)
Chemical product category	PC35: Washing and cleaning products (including solvent based products)
Environmental Release Categories	ERC8a: Wide dispersive indoor use of processing aids in open systems
Activity	Covers general exposures to consumers arising from washing and cleaning products

2.1 Contributing scenario controlling environmental exposure for: ERC8a

No exposure assessment presented for the environment.

2.2 Contributing scenario controlling consumer exposure for: PC35: Laundry handwashing

Product characteristics	Concentration of the Substance in Mixture/Article	Covers percentage substance in the product up to 5 %.
	Physical Form (at time of use)	liquid
	Vapour pressure	< 5 Pa
	Process Temperature	20 °C
Amount used	Amount used per event	7,8 g
Frequency and duration of use	Frequency of use	2 events/week
	Exposure duration per event	60 min
Human factors not influenced by risk management	Exposed skin areas	Fingertips 35,7 cm²
Other given operational conditions affecting consumers exposure	Room size	20 m³
	Covers use at ambient temperatures., Covers use under typical household ventilation.	
Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)	Consumer Measures	Wear suitable gloves.

2.3 Contributing scenario controlling consumer exposure for: PC35: Hand dishwashing liquids

Product characteristics	Concentration of the Substance in Mixture/Article	Covers percentage substance in the product up to 5 %.
	Physical Form (at time of use)	liquid
	Vapour pressure	< 5 Pa
	Process Temperature	20 °C

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Amount used	Amount used per event	3 g
Frequency and duration of use	Frequency of use	2 Times per day
	Exposure duration per event	60 min
Human factors not influenced by risk management	Exposed skin areas	Fingertips 35,7 cm ²
Other given operational conditions affecting consumers exposure	Room size	20 m ³
	Covers use at ambient temperatures., Covers use under typical household ventilation.	
Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)	Consumer Measures	Wear suitable gloves.

2.4 Contributing scenario controlling consumer exposure for: PC35: Surface cleaners (powder)

Product characteristics	Concentration of the Substance in Mixture/Article	Covers percentage substance in the product up to 1 %.
	Physical Form (at time of use)	solid
	Vapour pressure	< 5 Pa
	Process Temperature	20 °C
Amount used	Amount used per event	20 g
Frequency and duration of use	Frequency of use	2 Times per week
	Exposure duration per event	60 min
Human factors not influenced by risk management	Exposed skin areas	Fingertips 35,7 cm ²
Other given operational conditions affecting consumers exposure	Room size	20 m ³
	Covers use at ambient temperatures., Covers use under typical household ventilation.	
Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)	Consumer Measures	Wear suitable gloves.

2.5 Contributing scenario controlling consumer exposure for: PC35: Surface cleaners (spray)

Product characteristics	Concentration of the Substance in Mixture/Article	Covers percentage substance in the product up to 5 %.
	Physical Form (at time of use)	liquid
	Vapour pressure	< 5 Pa

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	Process Temperature	20 °C
Amount used	Amount used per event	5 g
Frequency and duration of use	Frequency of use	1 Times per week
	Exposure duration per event	60 min
Human factors not influenced by risk management	Exposed skin areas	Fingertips 35,7 cm ²
Other given operational conditions affecting consumers exposure	Room size	20 m ³
	Covers use at ambient temperatures., Covers use under typical household ventilation.	
Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)	Consumer Measures	Wear suitable gloves.

3. Exposure estimation and reference to its source

Consumers

Predicted exposures are not expected to exceed the applicable exposure limits when the operational conditions/risk management measures given in section 2 are implemented. The ECETOC TRA tool has been used to estimate consumer exposures unless otherwise indicated.

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

Health

The "Table of habits and practices for consumer products in Western Europe" developed by A.I.S.E. (2002) has been used to set the operational conditions as listed in section 2.2. The table can be found in the A.I.S.E. web site: http://www.aise.eu/reach/?page=exposureass_sub3

Where other Risk Management Measures/Operational Conditions are adopted, then users should ensure that risks are managed to at least equivalent levels.

For scaling see: <http://www.ecetoc.org/tra>

Only properly trained persons shall make use of scaling methods while checking whether the OC and RMM are within the boundaries set by the ES

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1. Short title of Exposure Scenario 5: Use in road and construction applications

Main User Groups	SU 22: Professional uses: Public domain (administration, education, entertainment, services, craftsmen)
Process categories	PROC8a: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)
Environmental Release Categories	ERC8c: Wide dispersive indoor use resulting in inclusion into or onto a matrix ERC8f: Wide dispersive outdoor use resulting in inclusion into or onto a matrix
Activity	Covers the use in constructions (application of concrete in construction activities)

2.1 Contributing scenario controlling environmental exposure for: ERC8c, ERC8f

No exposure assessment presented for the environment.

2.2 Contributing scenario controlling worker exposure for: PROC8a, PROC8b, PROC9

Product characteristics	Concentration of the Substance in Mixture/Article	Covers percentage substance in the product up to 100 % (unless stated differently).
	Physical Form (at time of use)	solid
	Vapour pressure	< 5 Pa
	Process Temperature	20 °C
Frequency and duration of use	Covers daily exposures up to 8 hours (unless stated differently).	
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training. (Efficiency: 90 %)(PROC8a)	
	Wear suitable gloves tested to EN374. (Efficiency: 80 %)(PROC8b, PROC9)	
	Wear a respirator conforming to EN 140/143 with type P1 filter or better (Efficiency: 80 %)(PROC8a, PROC8b, PROC9)	

3. Exposure estimation and reference to its source

Workers

Predicted exposures are not expected to exceed the applicable exposure limits when the operational conditions/risk management measures given in section 2 are implemented. The ECETOC TRA tool has been used to estimate workplace exposures unless otherwise indicated.

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

Health

Where other Risk Management Measures/Operational Conditions are adopted, then users should ensure that

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risks are managed to at least equivalent levels.

For scaling see: <http://www.ecetoc.org/tra>

Only properly trained persons shall make use of scaling methods while checking whether the OC and RMM are within the boundaries set by the ES

Additional good practice advice beyond the REACH Chemical Safety Assessment

Assumes a good basic standard of occupational hygiene is implemented.

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1. Short title of Exposure Scenario 6: Use in road and construction applications

Main User Groups	SU 21: Consumer uses: Private households (= general public = consumers)
Article categories	AC4: Stone, plaster, cement, glass and ceramic articles
Environmental Release Categories	ERC10a: Wide dispersive outdoor use of long-life articles and materials with low release ERC11a: Wide dispersive indoor use of long-life articles and materials with low release
Activity	Covers the use in construction (stone, plaster, cement)

2.1 Contributing scenario controlling environmental exposure for: ERC10a, ERC11a

No exposure assessment presented for the environment.

2.2 Contributing scenario controlling consumer exposure for: AC4

Product characteristics	Concentration of the Substance in Mixture/Article	Covers percentage substance in the product up to 1 %.
	Physical Form (at time of use)	solid
	Vapour pressure	< 5 Pa
	Process Temperature	20 °C
Amount used	Amount used per event	130 g
Frequency and duration of use	Frequency of use	4 days/year
	Frequency of use	1 Times per day
	Exposure duration per event	120 min
Human factors not influenced by risk management	Exposed skin areas	Covers skin contact area up to 1000 cm²
Other given operational conditions affecting consumers exposure	Room size	20 m³
	Covers use at ambient temperatures., Covers use under typical household ventilation.	
Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)	No specific risk management measure identified beyond those operational conditions stated.	

3. Exposure estimation and reference to its source

Environment

No exposure assessment presented for the environment.

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Consumers

Predicted exposures are not expected to exceed the applicable exposure limits when the operational conditions/risk management measures given in section 2 are implemented. The ECETOC TRA tool has been used to estimate consumer exposures unless otherwise indicated.

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario**Health**

Where other Risk Management Measures/Operational Conditions are adopted, then users should ensure that risks are managed to at least equivalent levels.

For scaling see: <http://www.ecetoc.org/tra>

Only properly trained persons shall make use of scaling methods while checking whether the OC and RMM are within the boundaries set by the ES

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1. Short title of Exposure Scenario 7: Use in glass and ceramics

Main User Groups	SU 22: Professional uses: Public domain (administration, education, entertainment, services, craftsmen)
Process categories	PROC8a: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)
Environmental Release Categories	ERC8c: Wide dispersive indoor use resulting in inclusion into or onto a matrix ERC8f: Wide dispersive outdoor use resulting in inclusion into or onto a matrix
Activity	Covers the application of ceramics in construction activities

2.1 Contributing scenario controlling environmental exposure for: ERC8c, ERC8f

No exposure assessment presented for the environment.

2.2 Contributing scenario controlling worker exposure for: PROC8a, PROC8b, PROC9

Product characteristics	Concentration of the Substance in Mixture/Article	Covers percentage substance in the product up to 100 % (unless stated differently).
	Physical Form (at time of use)	solid
	Vapour pressure	< 5 Pa
	Process Temperature	20 °C
Frequency and duration of use	Covers daily exposures up to 8 hours (unless stated differently).	
Technical conditions and measures to control dispersion from source towards the worker	Provide a good standard of general ventilation. Natural ventilation is from doors, windows etc. Controlled ventilation means air is supplied or removed by a powered fan.(PROC8a)	
Conditions and measures related to personal protection, hygiene and health evaluation	Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training. (Efficiency: 90 %)(PROC8a)	
	Wear suitable gloves tested to EN374. (Efficiency: 80 %)(PROC8b, PROC9)	
	Wear a respirator conforming to EN 140/143 with type P1 filter or better (Efficiency: 80 %)(PROC8b, PROC9)	

3. Exposure estimation and reference to its source

Workers

Predicted exposures are not expected to exceed the applicable exposure limits when the operational conditions/risk management measures given in section 2 are implemented. The ECETOC TRA tool has been used to estimate workplace exposures unless otherwise indicated.

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

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Health

Where other Risk Management Measures/Operational Conditions are adopted, then users should ensure that risks are managed to at least equivalent levels.

For scaling see: <http://www.ecetoc.org/tra>

Only properly trained persons shall make use of scaling methods while checking whether the OC and RMM are within the boundaries set by the ES

Additional good practice advice beyond the REACH Chemical Safety Assessment

Assumes a good basic standard of occupational hygiene is implemented.

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1. Short title of Exposure Scenario 8: Use in glass and ceramics

Main User Groups	SU 21: Consumer uses: Private households (= general public = consumers)
Article categories	AC4: Stone, plaster, cement, glass and ceramic articles
Environmental Release Categories	ERC10a: Wide dispersive outdoor use of long-life articles and materials with low release ERC11a: Wide dispersive indoor use of long-life articles and materials with low release
Activity	Covers general exposures to consumers arising from the use of ceramic tiles for flooring and walls

2.1 Contributing scenario controlling environmental exposure for: ERC10a, ERC11a

No exposure assessment presented for the environment.

2.2 Contributing scenario controlling consumer exposure for: AC4

Product characteristics	Concentration of the Substance in Mixture/Article	Covers percentage substance in the product up to 1 %.
	Physical Form (at time of use)	solid
	Vapour pressure	< 5 Pa
	Process Temperature	20 °C
Amount used	Amount used per event	1350 g
Frequency and duration of use	Frequency of use	3 events/year
	Frequency of use	1 Times per day
	Exposure duration per event	120 min
Human factors not influenced by risk management	Exposed skin areas	Covers skin contact area up to 1000 cm²
Other given operational conditions affecting consumers exposure	Room size	20 m³
	Covers use at ambient temperatures., Covers use under typical household ventilation.	
Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)	No specific risk management measure identified beyond those operational conditions stated.	

3. Exposure estimation and reference to its source

Consumers

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Predicted exposures are not expected to exceed the applicable exposure limits when the operational conditions/risk management measures given in section 2 are implemented. The ECETOC TRA tool has been used to estimate consumer exposures unless otherwise indicated.

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario**Health**

Where other Risk Management Measures/Operational Conditions are adopted, then users should ensure that risks are managed to at least equivalent levels.

For scaling see: <http://www.ecetoc.org/tra>

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